

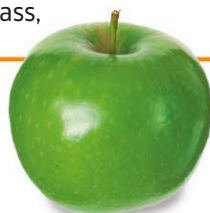
Derived quantities

Other kinds of quantities are also useful in science, but these are calculated from base quantities using scientific equations. For instance, we combine SI measurements of mass,

FORCE

SI unit: newton (N)

One newton is about the force of gravity on a single apple.



$$\text{Force in newtons} = \frac{\text{Mass in kilograms} \times \text{distance in meters}}{\text{Time in seconds}^2}$$

- A 10 billionth of a newton = the force needed to break six chemical bonds in a molecule.
- 10 newtons = the weight of an object with mass of 1 kilogram.

PRESSURE

SI unit: pascal (Pa)

One pascal is about the pressure of one bill of paper money resting on a flat surface.

$$\text{Pressure in pascals} = \frac{\text{Force in newtons}}{\text{Area in meters}^2}$$



- A 10 thousand trillionth of a pascal = the lowest pressure recorded in outer space.
- 1 million pascals (1 megapascal) = the pressure of a human bite.

ENERGY

SI unit: joule (J)

One joule is about the energy needed to lift a medium-sized tomato a height of one meter.



$$\text{Energy in joules} = \text{Force in newtons} \times \text{distance in meters}$$

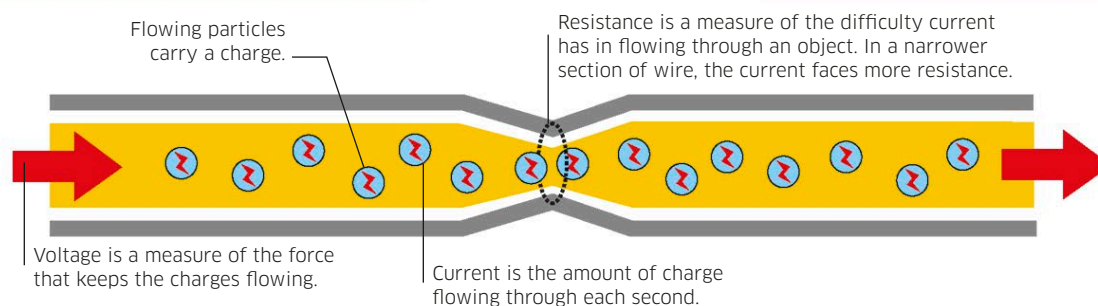
- A millionth of a joule (1 microjoule) = the energy of motion in six flying mosquitoes.
- 1,000 joules (1 kilojoule) = the maximum energy from the sun reaching 1 square meter of Earth's surface each second.

ELECTRICAL CHARGE AND RESISTANCE

SI unit: Charge—coulomb (C)

Resistance—ohm (Ω)

The measurements relating to electricity are all interlinked. Charge is a measure of how positive or negative particles are, and can be calculated from the current and the time. Resistance is a measure of the difficulty a current has in flowing, and can be calculated from the voltage and the current.



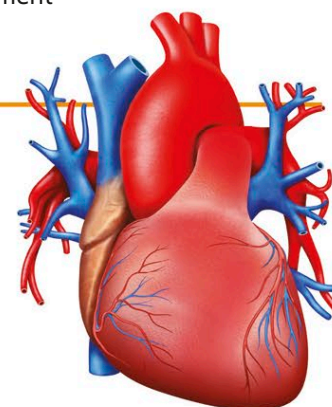
$$\text{Resistance in ohms} = \frac{\text{Potential difference in volts}}{\text{Current in amperes}}$$

distance, and time to work out an SI measurement for force. This means that force is said to be a derived quantity.

FREQUENCY

SI unit: hertz (Hz)

One hertz is about the frequency of a human heartbeat: one beat per second.



$$\text{Frequency in hertz} = \frac{\text{Number of cycles}}{\text{Time in seconds}}$$

- 100 hertz = the frequency of an engine cycle in a car running at maximum speed.
- 10,000 hertz = the frequency of radio waves.

POWER

SI unit: watt (W)

One watt is about the power used by a single Christmas tree light.



$$\text{Power in watts} = \frac{\text{Energy in joules}}{\text{Time in seconds}}$$

- A millionth of a watt (1 microwatt) = the power used by a wristwatch.
- 1 billion watts (1 gigawatt) = the power used by a hydroelectric generating station.

POTENTIAL DIFFERENCE

SI unit: volt (V)

Voltage is a measure of the difference in electrical energy between two points—the force needed to make electricity move. One volt is about the voltage in a lemon battery cell.



$$\text{Potential difference in volts} = \frac{\text{Power in watts}}{\text{Current in amperes}}$$

- 100 volts = the electrical grid voltage in the US.

$$\text{Charge in coulombs} = \text{Current in amperes} \times \text{time in seconds}$$